Controlled Ordinal Descents, Restricted Partitions and Cartier Observers

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Abstract

An ordinal descent can terminate at every input while its accumulated step count retains difficult modular information. We give exact count identities for a norm-controlled descent below ε_0. For a polynomial block, congruent growing budgets give the same count modulo a fixed modulus as a constant-budget restricted-partition problem. Applied to a genuinely retained prefix in the Wilken–Weiermann assignment, the identity produces a restricted-partition coefficient whose part bound and selected index both vary along one prescribed fast-growing orbit. We construct its complete finite coefficient observer. Differential Cartier removes the higher poles, but on the actual linked inputs the tuple of residues at one family of primitive roots has exact period $2(\log_2 B - 1)$. Thus a uniformly bounded cycle of the complete rational state cannot solve this family. The result does not determine the selected scalar coefficient, prove its nonperiodicity, or decide general canonical comparison. Complete counts, counts to a boundary, full states, and scalar observations are kept distinct throughout.

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1 Why a termination proof leaves a counting problem

Wilken and Weiermann associate typed terms with ordinal vectors whose zeroth coordinate is a natural number obtained from a norm-controlled collapse [1]. An exact evaluation follows an ordinal descent. Its ordinal argument decreases, but its available norm increases rapidly. The number of replacements is part of the answer, rather than merely a bound on the time needed to compute it.

This distinction matters for universal comparison. An algorithm that eventually evaluates each supplied input need not decide whether *some* input has a specified residue. Nor does a large growth rate determine a small residue. The methods below make that distinction explicit by following one actual source boundary all the way to a restricted-partition coefficient.

Our contribution has three levels. First, we prove an exact modular transfer theorem for polynomial blocks under congruent growing budgets. Second, we derive the correlated parameters of a block inside a nonpolynomial source descent. Third, we analyze a complete finite observer for its selected parity and prove an unbounded full-state period on those very parameters. The third result limits a particular compression strategy; it is not an impossibility theorem for all methods of observing the final bit.

2 Norm, control and the exact descent

All ordinals in this paper lie below ε_0 and are given by finite recursive Cantor normal forms. In displays of such forms, addition and multiplication by finite coefficients have their ordinary ordinal meaning. The norm is

$$\mathrm{N}(0) = 0, \qquad \mathrm{N}\left(\sum_{i<r} \omega^{\alpha_i} c_i\right) = \sum_{i<r} c_i(1 + \mathrm{N}(\alpha_i)).$$

Thus $\mathrm{N}(n) = n$, $\mathrm{N}(\omega^d) = d+1$ for finite d, and $\mathrm{N}(\omega^\omega) = 3$. Every norm ball is finite and effectively enumerable: its members are represented by finitely many rooted expression trees of bounded size, after which their Cantor forms can be compared effectively.

Fix exactly

$$F_0(n) = 2^n, \qquad F_{j+1}(n) = F_j^{n+1}(n), \qquad f(n) = \Phi(n) = F_5(n + 100). \tag{2.1}$$

Superscripts on functions denote iteration. In particular f is computable, strictly increasing, and

$$f(n) \geq 2^{n+100}, \qquad f(n+1) \geq 2^{f(n)+1}. \tag{2.2}$$

The finite-level iteration induction proving these growth estimates is given in the foundations companion [4]; the modular properties are proved below. The underlying collapse is

$$\psi(\alpha) = \max\big(\{0\} \cup \{\psi(\beta) + 1 : \beta < \alpha,\ \mathrm{N}(\beta) \leq f(\mathrm{N}(\alpha))\}\big).$$

The maximum is finite. It supplies the controlled implication

$$\beta < \alpha, \quad \mathrm{N}(\beta) \leq f(\mathrm{N}(\alpha)) \quad \Longrightarrow \quad \psi(\beta) + 1 \leq \psi(\alpha). \tag{2.3}$$

Unrestricted monotonicity of ψ is not asserted.

For $\alpha > 0$, let $M(\alpha, b) = \max\{\beta < \alpha : \mathrm{N}(\beta) \le b\}$. For a nonzero limit λ, its norm-saturating predecessor satisfies

$$\mathrm{N}(M(\lambda, b)) = b, \qquad \psi(\lambda + n) = \psi(M(\lambda, f(\mathrm{N}(\lambda) + n))) + 1. \tag{2.4}$$

These identities, including the terminating Cantor-form algorithm for M, are proved in the foundations companion [4]. They are a specialization of the norm-controlled predecessor framework of [2]. In particular, closed evaluation is an effective mathematical operation; no feasible running time is implied.

Definition 2.1. For a limit $\lambda > 0$ and total initial norm $t \ge \mathrm{N}(\lambda)$, start at $\alpha_0 = \lambda + (t - \mathrm{N}(\lambda))$. While α_i is infinite, discard its finite tail and set

$$\alpha_{i+1} = M(\lim(\alpha_i), f(\mathrm{N}(\alpha_i))).$$

Write $\tau_\lambda(t)$ for the number of replacements and $Z_\lambda(t)$ for the terminal finite ordinal. Set $J_\lambda(t) = \psi(\lambda + t - \mathrm{N}(\lambda))$.

Proposition 2.2 (Complete-count identity)**.** *The descent terminates and*

$$Z_\lambda(t) = f^{\tau_\lambda(t)}(t), \qquad J_\lambda(t) = Z_\lambda(t) + \tau_\lambda(t).$$

Proof. Every replacement strictly decreases the ordinal, so there cannot be infinitely many replacements. By norm saturation, its successive norms are exactly $t, f(t), f^2(t), \ldots$. At the finite endpoint, the ordinal itself equals its norm, proving the first identity. Each application of (2.4) adds one. Finally $\psi(n) = n$ for finite n, by induction in the defining maximum. Summing the contributions proves the second identity. □

This count stops at the *first finite ordinal*. It does not add a further walk through the finite successors below $Z_\lambda(t)$. Likewise, a count stopped at a nonfinite boundary is generally not τ_λ.

3 The modular information supplied by the control

We state the arithmetic input in a form that can be applied when a modulus itself depends on a source parameter.

Theorem 3.1 (Uniform residue envelopes)**.** *There is an effective function $m \mapsto r_m \in \{0, \ldots, m-1\}$ such that:*

1. *$f(x) \equiv r_m \pmod m$ for $x \ge m$;*
2. *for $s \ge 1$, $j \ge 2$ and $m \le 2^{2^s}$, $F_j(s) \equiv r_m \pmod m$;*
3. *if $B = f(u)$, then $f(u+1) \equiv r_m \pmod m$ for every $m \le 2^{2^B}$.*

Proof. We recall the construction to make the envelopes precise. Put $r_1 = 0$. For $m > 1$, write $m = 2^a b$, b odd, and set $q = \varphi(b)$ if $b > 1$, otherwise $q = 1$. Then $q < m$. Define r_m by $r_m \equiv 0 \pmod{2^a}$ and $r_m \equiv 2^{r_q} \pmod b$. This is a terminating recursion using Euler's theorem and the Chinese remainder theorem.

Let $E_0(n) = n$, $E_{h+1}(n) = 2^{E_h(n)}$. Induction on m proves that $E_h(n) \equiv r_m \pmod m$ whenever $h, n \geq m$. The exponent is large enough for the 2-part; its residue modulo q is supplied by the induction hypothesis, since $h-1 \geq q$. As $F_1(n) = E_{n+1}(n)$, every $F_j(n)$ with $j \geq 1$ has residue r_m for $n \geq m$: in a higher-level iterate every argument is at least n. This proves the first assertion.

For the stronger bound, the final F_1 call in $F_2(s) = F_1^{s+1}(s)$ has argument at least $F_1(s) \geq 2^{2^s}$ when $s \geq 1$. Its tower seed and height are both at least m, proving the second assertion for $j = 2$. The induction to higher j uses its last lower-level call, whose argument is at least s.

Finally put $n = u + 100$. In $F_5(n+1) = F_4^{n+2}(n+1)$ the argument of the final F_4 is $v = F_4^{n+1}(n+1) > F_4^{n+1}(n) = B$. The second assertion at v therefore gives the third. □

The theorem does not say that an arbitrary computable sequence of moduli may be stabilized at an arbitrary input. Each use must pay the displayed bound. In particular, a lower bound on $f(x)$ by itself is not the first assertion's lower bound on its *argument*.

4 Constant budgets and restricted partitions

For $d \geq 1$, define

$$p_d(n) = \#\{(a_1, \ldots, a_d) \in \mathbb{N}^d : a_1 + 2a_2 + \cdots + da_d = n\}.$$

The independent choices of multiplicities give

$$\sum_{n \geq 0} p_d(n) z^n = \prod_{i=1}^{d} (1 - z^i)^{-1}. \tag{4.1}$$

The relation to ordinals preserves every part, including the finite tail.

Lemma 4.1 (A fixed-budget count)**.** *Fix a nonnegative retained mass c and take $S \geq c + d + 1$. Start the suffix at ω^d with any finite tail of total norm (including c) at most S. At every replacement spend the constant suffix budget $S - c$ and stop when the suffix is finite. The number of replacements is $p_d(S - c)$. When $c = \mathrm{N}(\pi)$ for a retained CNF prefix π of least exponent at least d, this is precisely the corresponding segment starting at $\pi + \omega^d$ plus that tail.*

Proof. The ordinals of suffix norm exactly $S - c$ below ω^d are precisely

$$\omega^{d-1} a_d + \cdots + \omega a_2 + a_1, \qquad \sum_{i=1}^{d} i a_i = S - c. \tag{*}$$

Their number is $p_d(S - c)$. The first predecessor is the largest such ordinal. In any later state of (*), the nonfinite coefficients determine a_1 uniquely. Thus no second state of (*) has the same nonfinite part. Removing the finite tail and taking the largest predecessor of suffix norm $S - c$ selects exactly the next state of (*) in decreasing ordinal order. All states are visited, ending at the finite suffix $S - c$. There is one replacement into each state and none after the last, proving the count. For $d = 1$ there is one state and one replacement. □

Lemma 4.2 (A sufficient partition period)**.** *For every $d, m \geq 1$, put $Q_d = d! m^{d-1}$. Then*

$$p_d(n + Q_d) \equiv p_d(n) \pmod m \quad (n \geq 0).$$

Proof. For $d = 1$, $p_1(n) = 1$. For $d > 1$, splitting by the number of parts of size d gives $p_d(n) = \sum_{j=0}^{\lfloor n/d \rfloor} p_{d-1}(n - dj)$. Here $Q_d/d = mQ_{d-1}$. In the sum for $n + Q_d$, shifting the index by mQ_{d-1} recovers the old sum. The additional mQ_{d-1} summands form m groups of Q_{d-1} consecutive indices in the periodic sequence $j \mapsto p_{d-1}(n + Q_d - dj) \pmod m$. Its period divides Q_{d-1} by induction, so the extra sum is zero modulo m. □

The period is deliberately a sufficient bound, not an optimal one. For $d = 2$ the count is $\lfloor n/2 \rfloor + 1$, and $2m$ is the period supplied by the lemma.

5 A transfer theorem for congruent growing budgets

For the numerical statement, a retained mass $c \geq 0$ means that a suffix of norm n has total norm $c+n$, and a total budget b leaves suffix budget $b-c$. It need not be the norm of an ordinal prefix. If it is realized by an actual retained CNF prefix, the numerical process agrees with that source segment. A budget schedule may increase with the current total norm; it must always afford that norm.

Theorem 5.1 (Congruent-budget transfer)**.** *Consider deletion of one suffix block ω^d, with retained numerical mass $c \geq 0$ and initial total norm at least $c + d + 1$, under budgets at least the current total norm. Suppose every budget used in this segment is congruent to one integer S modulo $Q_d = d!m^{d-1}$, where $S \geq c$. Then its complete segment count is congruent to $p_d(S - c)$ modulo m.*

Proof. Induct on d. A degree-one block takes one replacement, as does $p_1(S - c)$. For the induction step let b be the first actual total budget. The suffix of its greatest predecessor is the greedy Cantor polynomial of norm $b - c$ and degree less than d. Write its coefficients as $a_{d-1}, \ldots, a_1$, with the final finite coefficient included.

Thereafter each of its nonfinite monomial units is deleted by a lower-degree block. For a degree-i unit the retained prefix norm has the form

$$c_i = c + \sum_{j>i} (j+1)a_j + (i+1)a, \qquad 0 \leq a < a_i. \tag{**}$$

The order of those units is fixed by the Cantor order; growing budgets affect their lower suffixes and durations, not the number of these fixed higher monomial units. Since $Q_i \mid Q_d$, every budget in a child segment has the residue of b modulo Q_i. Induction makes its count congruent to $p_i(b - c_i)$. These arguments are nonnegative: each c_i consists of c plus the norm of a prefix of the first suffix polynomial and is at most b.

Consequently the whole count, including its initial replacement, has the same residue as the constant-b process with exactly that initial polynomial and those prefixes (**). By Lemma 4.1 this is $p_d(b - c)$. Since $b \equiv S \pmod{Q_d}$, Lemma 4.2 gives $p_d(b - c) \equiv p_d(S - c) \pmod m$. If $b - c < S - c$, apply the period lemma in the reverse direction by adding the appropriate nonnegative multiple; no negative partition index is used. □

The proof covers the whole count, not just its leading contribution. It does not replace the increasing operational budgets by a constant control: the latter is only a device for computing one residue.

Proposition 5.2 (An actual ω^ω block)**.** *Let $\Omega = \omega^\omega$. At a real suffix state $\pi + \Omega + n$, with π a retained CNF prefix (zero or of least exponent at least ω), and total norm u, assume $c = \mathrm{N}(\pi)$, $c + 3 \le u$, and $B = f(u) \ge \max(16, m)$. Put $C = f(u+1)$ and $d = B - c - 1$. If $H_c(u)$ counts the complete deletion of this Ω copy, then*

$$H_c(u) \equiv 1 + p_{B-c-1}(C - c) \pmod{m}. \tag{5.1}$$

Its output norm is still exactly $f^{H_c(u)}(u)$.

Proof. The first predecessor is $\pi + \omega^d$, of total norm B. Indeed the greatest exponent of norm at most $B - c - 1$ is the finite exponent d in this particular bound below ω; this is the predecessor of ω^ω under the suffix budget. All subsequent norms are at least B, and their budgets have residue r_{Q_d} by Theorem 3.1, because

$$Q_d = d! m^{d-1} \le B^{2B} \le 2^{2^B}.$$

The last inequality holds for $B \ge 16$; it follows from $2B \log_2 B \le 2^B$, first at 16 and then by the increasing ratio of the right side to the left side. The adjacent value $C = f(u+1)$ has the same residue modulo Q_d by the third part of Theorem 3.1. Apply Theorem 5.1 to the remaining ω^d block with $S = C$, then include its preceding Ω replacement. Norm saturation proves the final assertion independently of the modular calculation. □

6 The first moving-prefix boundary

Write $\Theta = \omega^{\omega+1}$. Suppose a genuine descent first reaches a state whose limit part is Θ, at total norm v. The forthcoming calculation is conditional only on this actual entry; it does not posit independent entry values for a larger source. Put $u = f(v)$. The residue recursion gives $u \equiv 1 \pmod 3$.

Lemma 6.1 (The retained prefix at first entry)**.** *The predecessor of this Θ state is*

$$\Omega q + 1, \qquad q = (u-1)/3.$$

The first Ω copy deleted in this polynomial has retained prefix norm $e = u - 4$ and total input norm u.

Proof. A copy of Ω has norm three. The maximal exponent below $\omega + 1$ that can occur is ω; the remaining budget is therefore spent on as many copies of Ω as possible and then a finite remainder. Since $u \equiv 1 \pmod 3$, that remainder is one. Before deleting the last displayed Ω in the ordinary sum $\Omega(q-1) + (\Omega + 1)$, the retained prefix is $\Omega(q-1)$ and has norm $3(q-1) = u - 4$. □

With $B = f(u)$, $C = f(u+1)$, $e = u - 4$ and $d = B - u + 3$, the first block parity is consequently

$$\delta_4(v) := H_{u-4}(u) \bmod 2 = 1 + p_d(C - e) \pmod 2. \tag{6.1}$$

The subscript records the norm four of its suffix $\Omega + 1$. The initial Θ replacement lies outside $H_{u-4}(u)$; the displayed 1 instead counts the Ω replacement.

This quantity is inside a larger correlated computation. Before its remaining limit part has become exactly Ω, the Θ segment deletes $q - 1$ copies with prefix masses $3(q-1), 3(q-2), \ldots, 3$. Their inputs are successive points on the *same* f orbit. Observing the sum of all these counts does

not automatically define the graph of the first summand. Thus (6.1) is a concrete benchmark for a proposed arithmetic method, not a proved necessary subproblem of every possible comparison procedure.

7 The complete binary coefficient observer

Work over $\mathbb{F}_2$ and set

$$A_d(z) = \prod_{i=1}^{d} (1 + z^i), \qquad D = d(d+1)/2.$$

For $b \in \{0, 1\}$ let $\Lambda_b(\sum a_n z^n) = \sum a_{2n+b} z^n$. The state space of proper numerators is $V_d = \{U \in \mathbb{F}_2[z] : \deg U < D\}$.

Lemma 7.1 (Exact state update)**.** *For $U \in V_d$,*

$$\Lambda_b(U/A_d) = \frac{\Lambda_b(UA_d)}{A_d}.$$

The updated numerator is again in V_d. In the coefficient basis its matrix is $T_b[i, j] = [z^{2i+b-j}]A_d$, for $0 \le i, j < D$.

Proof. Since $A_d(z)^2 = A_d(z^2)$ in characteristic two, $U/A_d = (UA_d)/A_d(z^2)$. Coefficient selection commutes with a factor in z^2, giving the identity. The numerator degree is at most $\lfloor (2D-1)/2 \rfloor = D - 1$. Applying selection to each monomial basis vector gives the matrix entries. □

Theorem 7.2 (Complete selected coefficient)**.** *If $n = \sum_{i<L} b_i 2^i$, with bits read least significant first, then*

$$p_d(n) \bmod 2 = e_0^T T_{b_{L-1}} \cdots T_{b_0} e_0.$$

Proof. The initial numerator is one, represented by e_0. Successive selections with $b_0, b_1, \ldots$ discard these bits of the coefficient index. After all bits, the desired coefficient is the constant term. As $A_d(0) = 1$, this is the constant numerator coordinate selected by e_0^T. □

This formula retains all factors and all cancellations. Boolean reachability in its transition graph would be a different observation: two paths contributing to one coefficient cancel in $\mathbb{F}_2$.

Lemma 7.3 (All-continuation observability)**.** *For any field K, polynomial A with $A(0) = 1$ and $\deg A = D$, the first D coefficients of U/A determine every U of degree less than D. For the binary observer above, distinct numerator states are distinguished by some continuation of length at most $\lceil \log_2 D \rceil$.*

Proof. Multiplication by $1/A = 1 + a_1 z + \cdots$ sends the coefficient vector of U to the first D output coefficients by a lower triangular matrix with diagonal one. It is invertible. Equivalently, the first nonzero coefficient of a difference $U - V$ remains its first nonzero coefficient after division by A. Its index $k < D$ has at most $\lceil \log_2 D \rceil$ bits; those bits form a distinguishing continuation. Index zero permits the empty word. □

This is the linear observability viewpoint familiar from weighted automata [3]. It does not say that every distinguishing continuation is available from the genuine source parameters. A quotient preserving only the one prescribed source word remains a different question.

8 Differential Cartier and the surviving residues

It is useful to normalize the same selected coefficient as a differential. Over a perfect field K of characteristic two, every rational function has a unique expression $a(z)^2 + zb(z)^2$. Define $\mathcal{C}((a^2 + zb^2)\,dz) = b\,dz$. This is the differential Cartier operator. Multiplication by a square pulls out its square root.

Lemma 8.1 (Local rule)**.** *For a fixed pole ζ and $b \in K$,*

$$\mathcal{C}\left(\frac{b\,dz}{(z-\zeta)^{2j}}\right) = 0, \qquad \mathcal{C}\left(\frac{b\,dz}{(z-\zeta)^{2j+1}}\right) = \frac{b^{1/2}\,dz}{(z-\zeta)^{j+1}}.$$

The same definition gives $\mathcal{C}(\sum a_n z^n\,dz) = \sum a_{2n+1}^{1/2} z^n\,dz$.

Proof. An even power of $h = z - \zeta$ is a square and contributes zero. For odd powers, write $h^{2k+1} = (h^k)^2(z-\zeta)$. The constant differential $\zeta\,dz$ has Cartier image zero, whereas $\mathcal{C}(z\,dz) = dz$. Pulling out the square proves the rule for all integer k, including negative k. The coefficient rule follows by the same even/odd decomposition. □

For $e \geq 1$ and $2^\ell \geq e$,

$$[z^0\,dz]\,\mathcal{C}^\ell\left(\frac{z^{e-1}}{A_d(z)}\,dz\right) = p_d(2^\ell - e) \pmod 2. \tag{8.1}$$

Indeed ℓ odd selections choose original exponent $2^\ell - 1$, which in z^{e-1}/A_d has coefficient $p_d(2^\ell - e)$.

Lemma 8.2 (Removal of every higher pole)**.** *Put $\mu = \sum_{i=1}^{d} 2^{\nu_2(i)}$ and $a = \lceil \log_2 \mu \rceil$. After a Cartier applications to a proper differential with denominator A_d, all surviving finite poles are simple. Its simple residues subsequently undergo inverse Frobenius at their fixed poles. No polynomial part is created.*

Proof. The multiplicity at a root of odd order q is $\sum_{q|i,\ i\leq d} 2^{\nu_2(i)} \leq \mu$. By Lemma 8.1, a term of order $k > 1$ survives a applications only if $k - 1$ is a positive multiple of 2^a. This is impossible for $k \leq \mu \leq 2^a$. A simple residue is replaced by its square root. The proper partial-fraction expansion has no polynomial part, and every term in its Cartier image is again proper, proving the final assertion. □

Here $\nu_2(i)$ denotes the exponent of two in the positive integer i. The simple-pole tail is indeed eventually periodic, rather than merely subject to a conditional period bound: all its finitely many poles and residues lie in one finite extension of $\mathbb{F}_2$, and inverse Frobenius is a permutation of that finite field. This establishes a finite cycle for every supplied cutoff, without giving a bound independent of the cutoff.

For the linked values in (6.1), $C = 2^\ell$ with $\ell \geq B + 1$ by (2.2). Since $\mu \leq d(1 + \lfloor \log_2 d \rfloor) \leq B(1 + \log_2 B)$, we have $a \leq B$ for $B \geq 16$. Thus the actual selected observation is beyond the higher-pole transient. This pays an actual parameter bound; it is not an unspecified promise that enough iterations occur.

The complete state can nevertheless have a long simple-pole cycle. Pole count alone cannot prove that. For example, dS/S for a squarefree polynomial S has residue one at every simple pole and is Cartier-fixed, however many poles S has.

9 An exact period on the actual linked inputs

Retain u, B, C, e, d from (6.1), and write

$$B = 2^K, \qquad r = K - 1, \qquad q = 2^r + 1 = B/2 + 1, \qquad m_0 = u - 2, \qquad \Delta = m_0(m_0+1)/2.$$

The integer q need not be prime. All primitive q-th roots in an algebraic closure of $\mathbb{F}_2$ will be used.

Lemma 9.1 (The actual residue formula)**.** *The form $\omega_e = z^{e-1} A_d(z)^{-1}\, dz$ has a simple nonzero residue at each primitive q-th root ζ, given by*

$$b_\zeta = \frac{\zeta^e}{A_{d-q}(\zeta)} = \zeta^{e-\Delta} A_{m_0}(\zeta).$$

Proof. The hierarchy bounds give $d/2 < q \le d < 2q$ and $m_0 < q$. Thus only $1 + z^q$ among the factors of A_d vanishes at ζ, and its derivative there is ζ^{-1}. The product $\prod_{i=1}^{q-1}(1+\zeta^i)$ is one: it is the derivative at one of $X^q - 1$, whose value is $q = 1$ in characteristic two. It follows that the residue is the first displayed expression.

Now $d - q = q - 1 - m_0$. The last m_0 terms in that complete product equal $\prod_{j=1}^{m_0}(1+\zeta^{-j}) = \zeta^{-\Delta} A_{m_0}(\zeta)$. Taking the complementary product proves the second expression. None of $1+\zeta^j$ for $1 \le j \le m_0$ vanishes. □

Lemma 9.2 (The bounds needed for the root count)**.** *For the actual linked inputs, $r \ge 96$, $\Delta < r$, and*

$$0 < \Delta - 2e < q, \qquad \varphi(q) > \frac{2^r}{r+1} > 2^{2r/3}\Delta.$$

Proof. For $s = u+100$, $F_5(s) \ge F_1(s) \ge 2^{2^s}$, so $K \ge 2^{u+100}$ and $r \ge 2^{u+100} - 1$. These inequalities imply $r \ge 96$ and $\Delta = (u-2)(u-1)/2 < r$. As $e = u - 4 = m_0 - 2$, $\Delta - 2e = (m_0^2 - 3m_0 + 8)/2$ is positive and smaller than q by the same growth bound.

If an odd integer q has k distinct prime factors $p_1 < \cdots < p_k$, then $p_i \ge i + 2$, hence $q/\varphi(q) = \prod p_i/(p_i - 1) \le k+1$. Here $3^k \le q = 2^r + 1$ implies $k \le r$, so $\varphi(q) \ge q/(r+1) > 2^r/(r+1)$. Finally $2^{r/3} > r^3$ at $r = 96$ and thereafter: the ratio of the exponential to r^3 increases for $r \ge 96$. Since $\Delta < r$ and $r(r+1) < r^3$, this gives $2^r/(r+1) > 2^{2r/3}\Delta$. □

Theorem 9.3 (Exact actual residue-tuple period)**.** *The tuple $(b_\zeta)_{\zeta \text{ primitive } q}$ has exact Frobenius period $2r$. Every eventual period of the complete rational differential Cartier orbit of ω_e is divisible by $2r = 2(\log_2 B - 1)$.*

Proof. The order of two modulo $q = 2^r + 1$ is $2r$. It divides $2r$ because $2^r \equiv -1 \pmod q$. Any proper divisor of $2r$ is at most r, and for $s \le r$ the positive integer $2^s - 1$ is smaller than q. Thus no smaller order is possible. The primitive roots and their residues belong to $\mathbb{F}_{2^{2r}}$, so the tuple period divides $2r$.

Reciprocity gives $A_{m_0}(\zeta^{-1}) = \zeta^{-\Delta} A_{m_0}(\zeta)$. If Frobenius power r fixed the tuple, Lemma 9.1 would give $\zeta^{\Delta - 2e} = 1$ at a primitive root, contrary to $0 < \Delta - 2e < q$.

Any other proper divisor s of $2r$ is at most $2r/3$: the integer quotient $(2r)/s$ is then at least three. Fixedness at all primitive roots would make every such root a zero of

$$P_s(X) = A_{m_0}(X)^{2^s} + X^{(\Delta - e)(2^s - 1)} A_{m_0}(X).$$

This polynomial has constant coefficient one and degree exactly $2^s\Delta$, since $e > 0$ makes the second summand's degree strictly smaller. It cannot have all $\varphi(q)$ primitive roots as zeros, by Lemma 9.2. Thus no proper divisor is a tuple period.

The chosen finite poles were already simple initially. By Lemma 8.1, the tuple in the j-th Cartier state is $(b_\zeta^{2^{-j}})_\zeta$ with poles fixed. If states at j and $j+p$ agree, these tuples agree. Inverse Frobenius is bijective and its tuple orbit has period $2r$, so $2r \mid p$. This applies also after the higher-pole transient. □

The proof is a tuple argument. It does not assert that every individual residue has degree $2r$, and it does not calculate the selected constant coefficient. In particular, the growing tuple period does not prove that the sequence $v \mapsto \delta_4(v)$ is nonperiodic.

10 What has and has not been reduced

For each actual input, the observer is finite and complete. Its size grows with d, and its exact rational tail has no uniformly bounded state period. These facts explain why a proposed solution must do more than attach a finite automaton to each supplied cutoff.

One possible improvement would preserve only the scalar observations allowed by the linked source inputs. The all-continuation Lemma 7.3 leaves that possibility open: it distinguishes states using arbitrary binary continuations, not just the prescribed source word. Another possibility would classify complete descent-count residues without first evaluating the isolated boundary bit. Neither possibility is supplied by the present arguments.

There are therefore three distinct mathematical outcomes here: an exact congruence transferring a growing-budget count to a restricted partition; a complete finite observer for that coefficient; and an exact full-state period obstruction on genuine inputs. Their proofs do not settle the remaining fixed-control all-ground canonical comparison problem. The source-expressivity companion [5] explains why complete-count attainment is an actual necessary arithmetic problem, while extracting one summand from the complete count requires an additional interpretation.

References

[1] Gunnar Wilken and Andreas Weiermann. Derivation lengths classification of Gödel's T extending Howard's assignment. *Logical Methods in Computer Science* **8**(1:19) (2012), 1–44. doi:10.2168/LMCS-8(1:19)2012. Official article and full text. The 2012 revised article is the version used throughout.

[2] W. Buchholz, A. Cichon and A. Weiermann. *A Uniform Approach to Fundamental Sequences and Hierarchies.* Mathematical Logic Quarterly **40**(2) (1994), 273–286. doi:10.1002/malq.19940400212. The norm-defined predecessor framework is discussed on p. 274; the additive symbol-count norm appears on p. 283.

[3] Stefan Kiefer. *Notes on equivalence and minimization of weighted automata.* arXiv:2009.01217v1, 2 September 2020. doi:10.48550/arXiv.2009.01217. The supplied first version is the version inspected.

[4] *The Wilken–Weiermann Comparison Problem: Foundations and a Guide to the Results.* Companion working manuscript, 2026. Included manuscript.

[5] *Actual-Source Arithmetic and Undecidability Boundaries for Canonical Comparison.* Companion working manuscript, 2026. Included manuscript.